



Department of Computer Science and Engineering (Data Science)

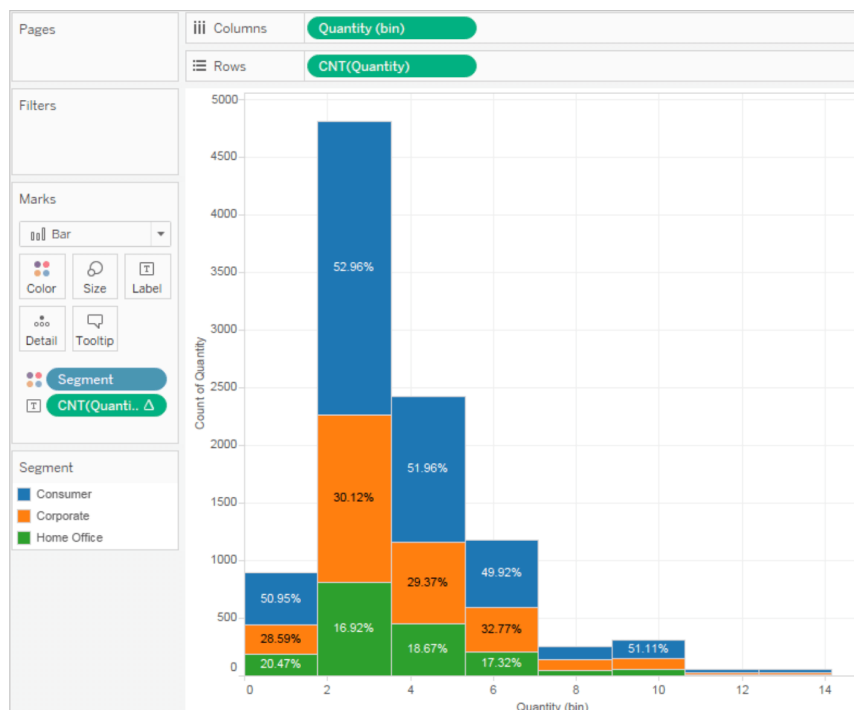
Experiment 5 (Data Visualization)

Aim: Perform bins, distributions and create heat maps on a given dataset.

Theory:

Bins:

Any discrete field in Tableau can be considered as a set of bins. For example, suppose you create a view with **Profit** on **Rows** and **State** on **Columns**. You could consider the **State** field as a set of bins—each profit value is sorted into a bin corresponding to the state from which the value was recorded. But if you want to see values for **Profit** assigned to bins without reference to a dimension, you can create a numeric bin, with each individual bin corresponding to a range of values.



You can bin data only for relational data sources and binned fields cannot be used in calculations. However, it is possible to create a calculated field that will replicate a bin of a specific size. For example: $(\text{FLOOR}([\text{Sales}]/1000) * 1000)$ will create bins with a size of 1000. By dragging this calculation to the dimension's pane, you can use these bins with cube data sources and calculated fields.

Heat Maps:



Department of Computer Science and Engineering (Data Science)

A heat map is a two-dimensional representation of data in which values are represented by colors. A simple heat map provides an immediate visual summary of information. More elaborate heat maps allow the viewer to understand complex data sets.

There can be many ways to display heat maps, but they all share one thing in common -- they use color to communicate relationships between data values that would be much harder to understand if presented numerically in a spreadsheet.

In the United States, many people are familiar with heat maps from viewing television news programs. During a presidential election, for instance, a geographic heat map with the colors red and blue will quickly inform the viewer which states each candidate has won.



Lab Assignments to complete in this session

1. Use the given datasets and perform the following tasks:

Dataset:

https://data-flair.training/blogs/download-electronic-store-sales-data/?ref=morioh.com&utm_source=morioh.com

The historical data contains tons of records for the electronic products sale. The stores are located in different regions, and each store includes several segments. Create bins on sales measure and visualize the data and filter the same using regions.

2. Use the given datasets and perform the following tasks:

Dataset:

<https://data.world/annjackson/world-indicators-tableau>

The UN also has an Advisory Role. It advises countries on things that require improvement/change. Use a Highlight table to find if any of the countries with a large population require changes.

The Highlight table would be used to compare the parameters with other countries in order to evaluate if anomalies are present.

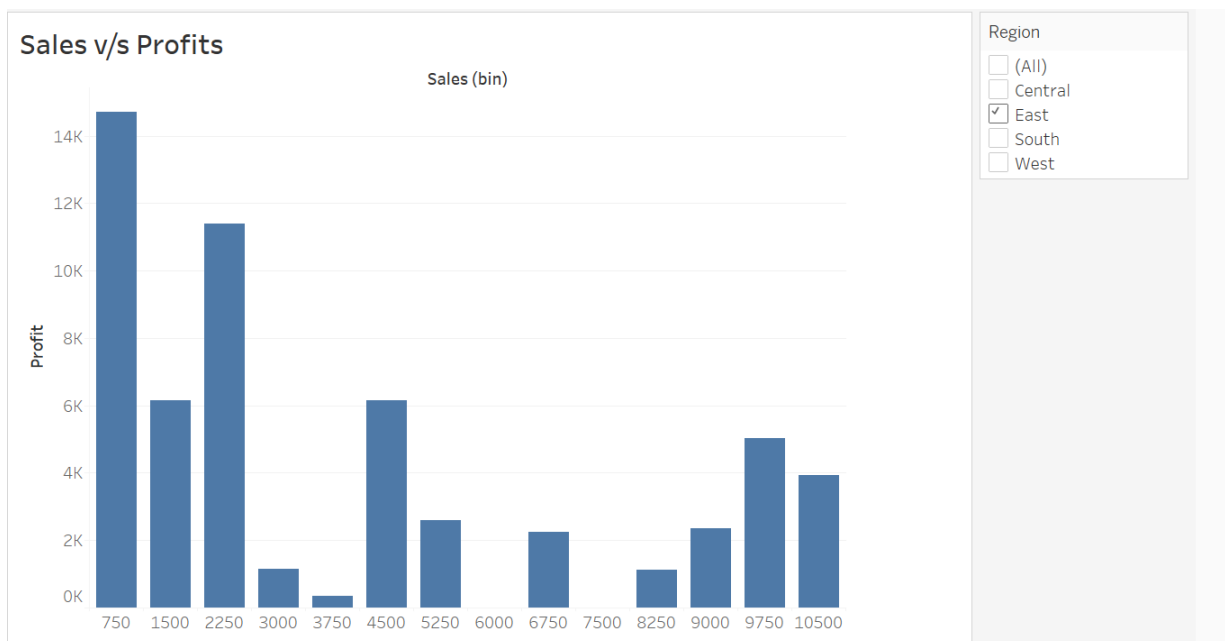
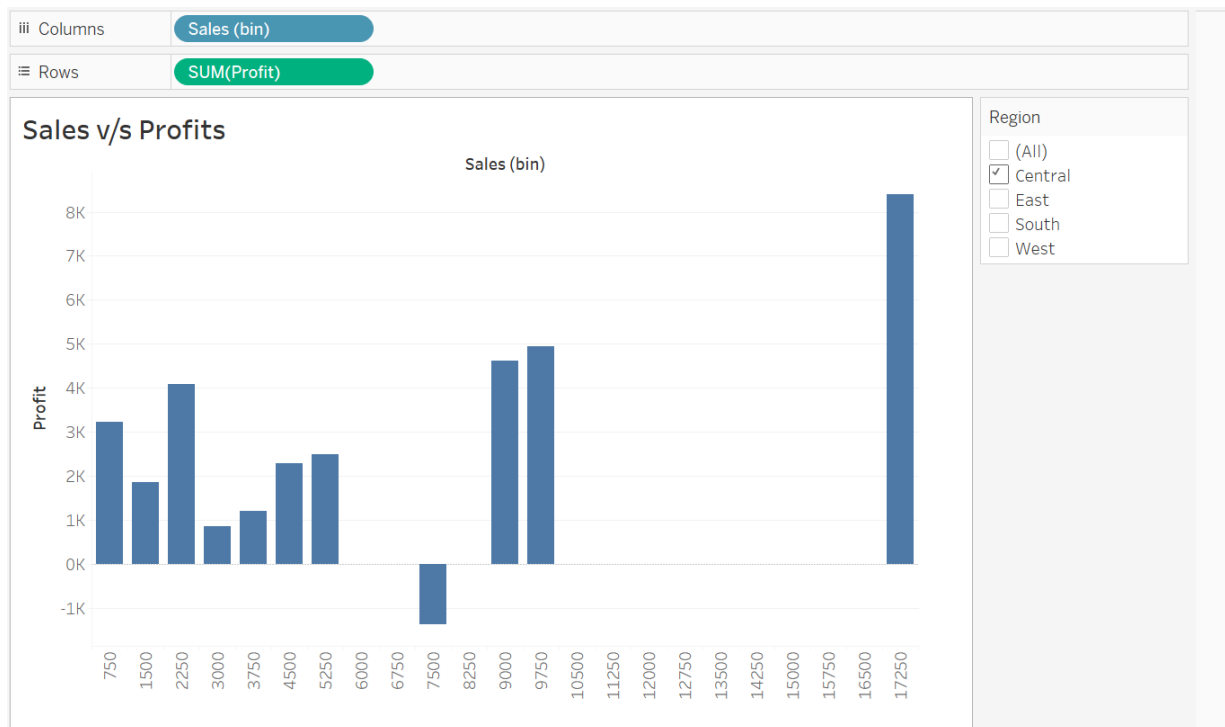
Health and/or Development Metrics should be used for the same

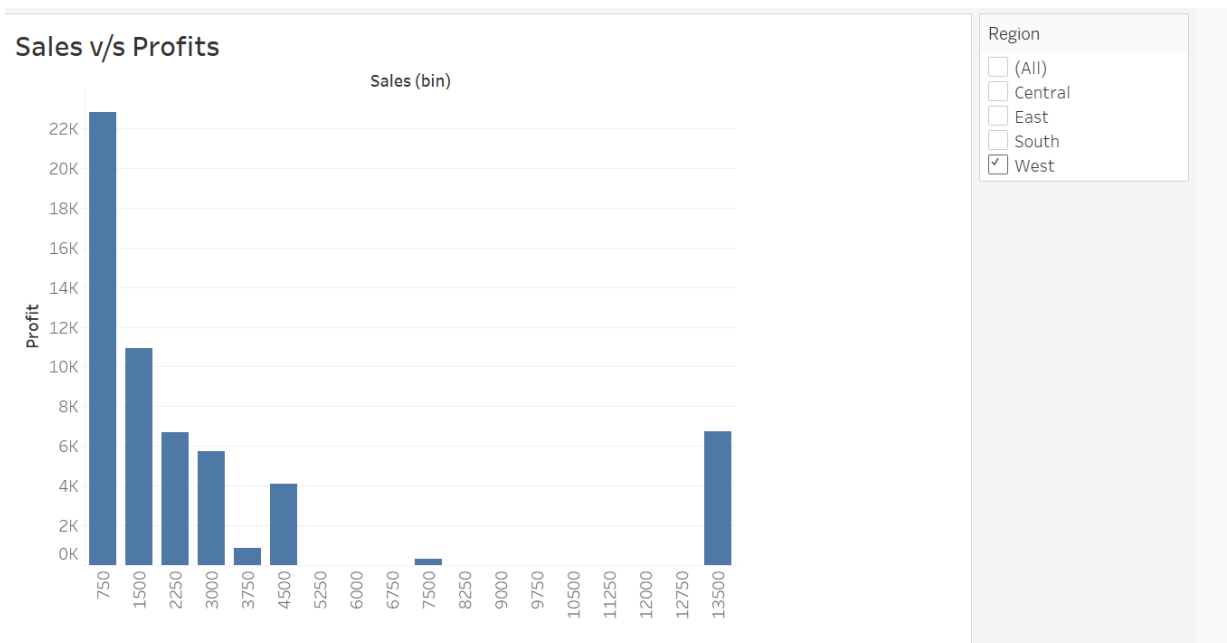
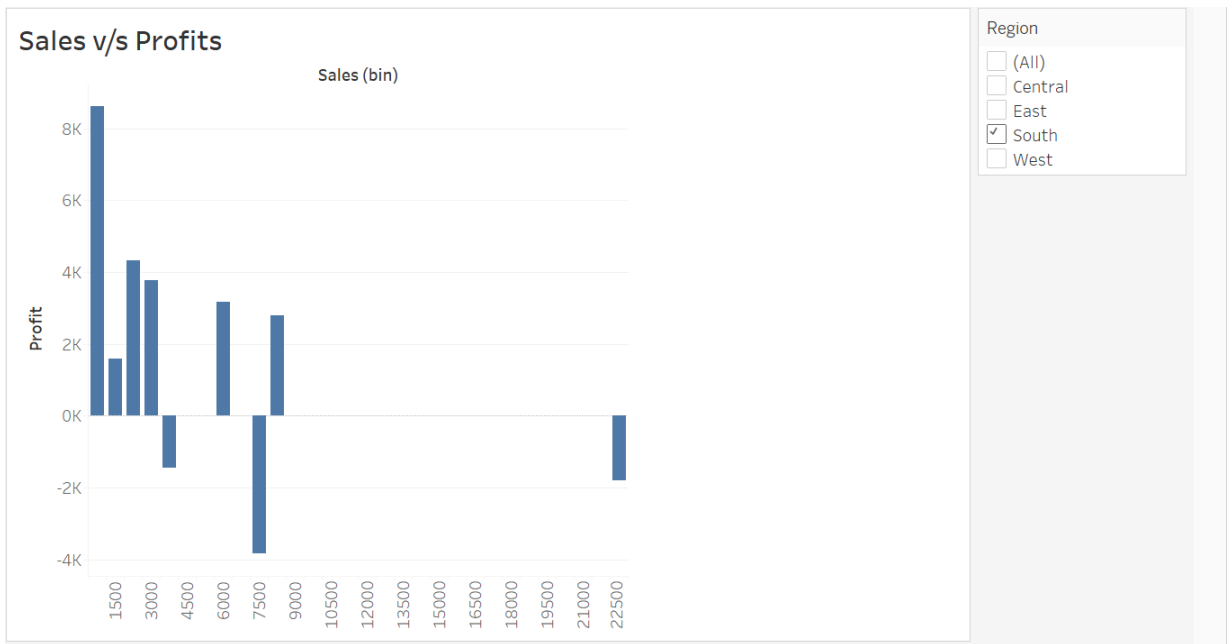
FDA Experiment-5

Name: Bhuvi Ghosh
SAPID:60009210191

Solution:-

The following are the region-wise visualizations for sales v/s profits:

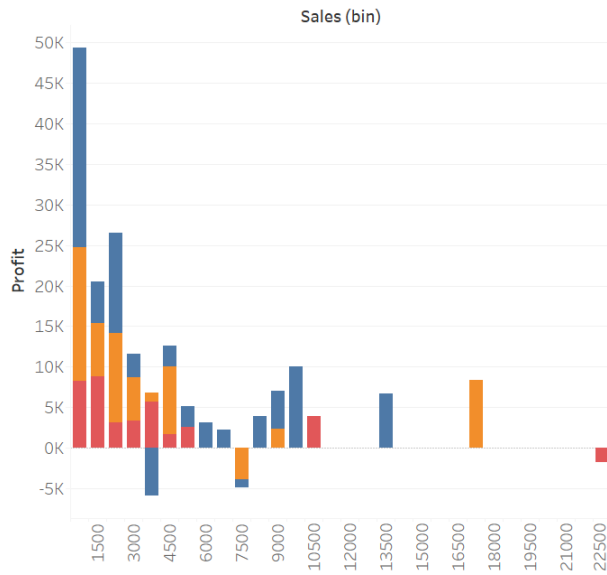




It is clearly evident that the southern region has suffered the maximum losses for sales ranging between 6,000 – 7,500 and 21,000 – 22,500.

The following are the region-wise and segment-wise visualizations for sales v/s profits:

Sales v/s Profits



Region

☒ (All)

☒ Central

☒ East

☒ South

☒ West

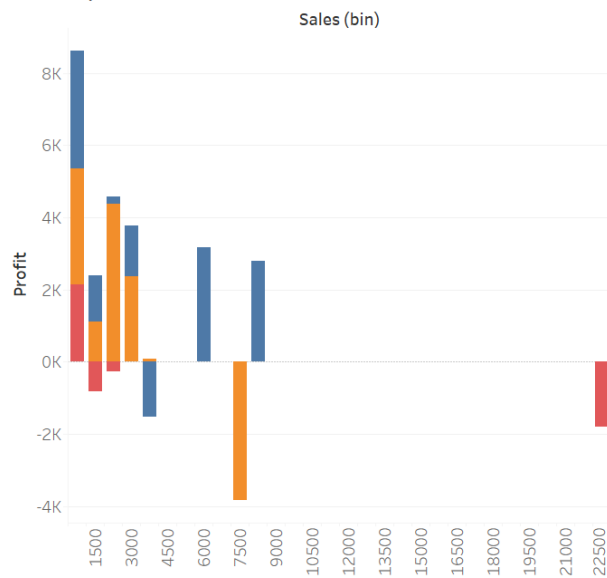
Segment

☒ Consumer

☒ Corporate

☒ Home Office

Sales v/s Profits



Region

☐ (All)

☐ Central

☐ East

☒ South

☐ West

Segment

☒ Consumer

☒ Corporate

☒ Home Office

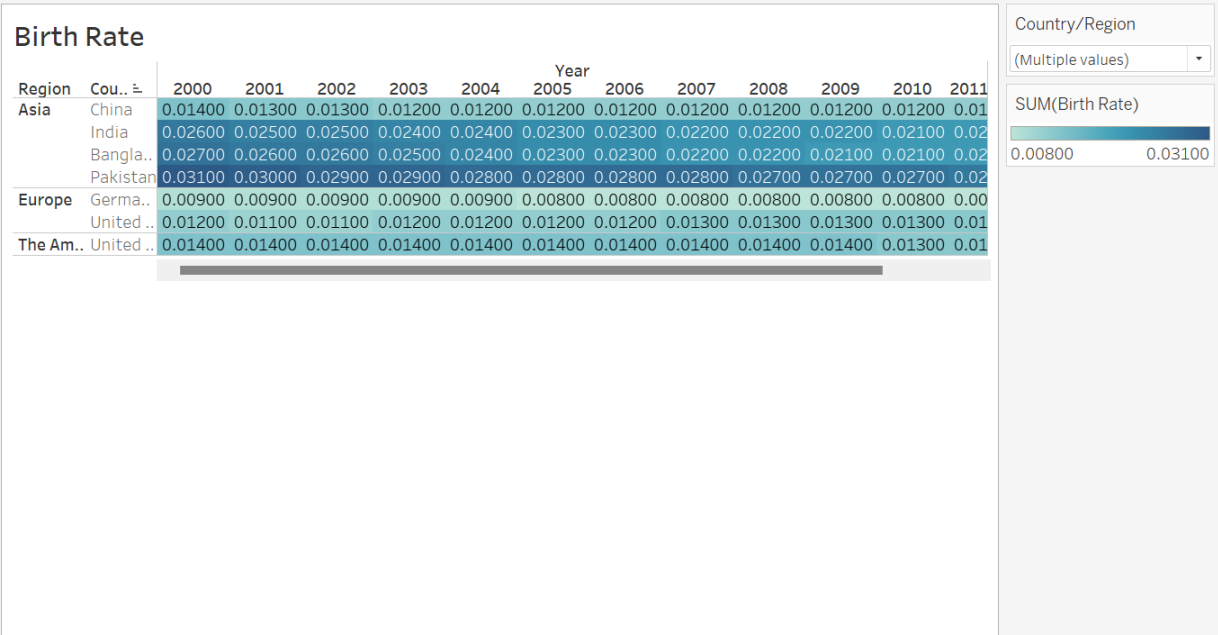
Thus, it can be concluded that the significant losses in the southern region have been sustained in the corporate, and home office sectors, with the former being the most loss making segment.

Topic: UN Advisory Role

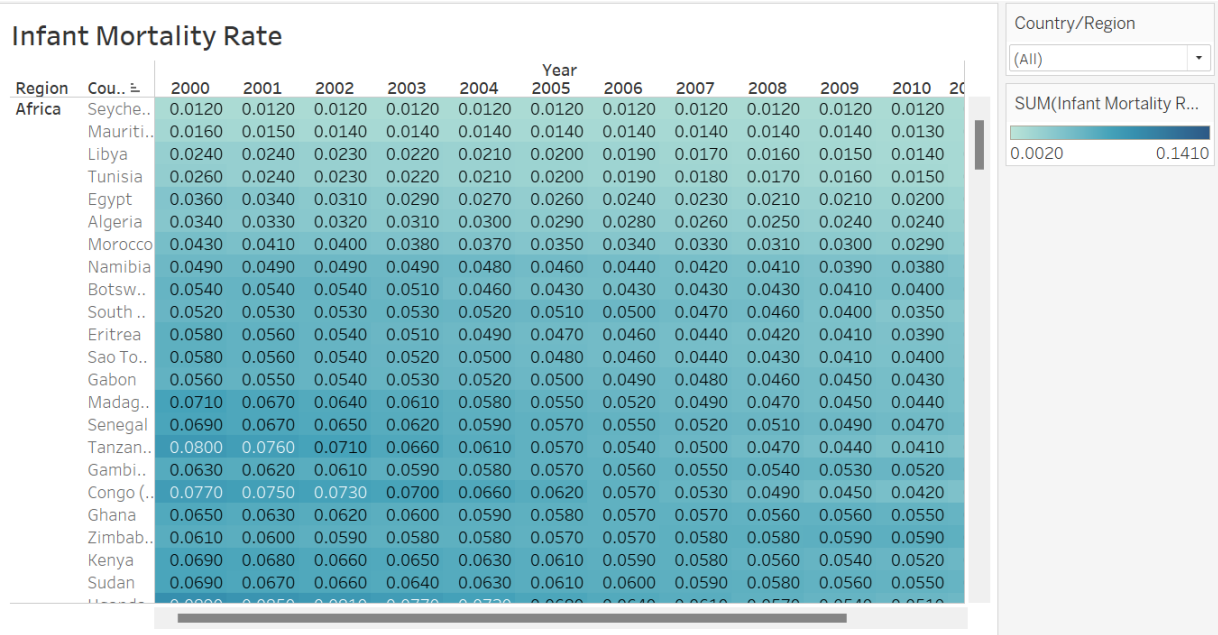
Solution:-

Health Advisory Metrics:

- Birth Rate:** The birth rate has been higher in general for Asian countries as compared to European and American countries. Pakistan seems to have the highest birth rate, throughout the years, among the countries compared below.



- Infant Mortality Rate:** Low for African countries in general.



- **Health Expenses/ Capita:** The United States has the highest health expenses per capita, whereas in Africa, South Africa has much higher values than Central African Republic. Both India and China have lower values for this metric, however, China's is much more than India.

Region	Country/Region	Year												
		2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Africa	South Africa	246	223	205	310	410	450	455	449	437	484	615	670	645
	Central African R..	11	10	11	13	14	15	15	18	20	17	18	19	18
Asia	China	43	47	54	61	70	80	93	113	155	189	216	274	322
	India	20	21	21	24	29	32	33	40	43	44	52	62	61
Europe	United Kingdom	1,761	1,819	2,065	2,444	2,955	3,161	3,423	3,953	3,875	3,512	3,489	3,659	3,647
The Americas	United States	4,790	5,138	5,578	5,993	6,355	6,732	7,110	7,486	7,769	8,009	8,254	8,467	8,895

Country/Region
(Multiple values)
SUM(Health Exp/Capita)
10 8,895

Development Metrics:

- **GDP:** The top 5 economies with highest GDP in Asia, from 2000 to 2004 were, Japan, China, India, South Korea and Indonesia.

Region	Country	Year				
		2000	2001	2002	2003	2004
Asia	Japan	47,30,00,00,00,000	41,60,00,00,00,000	39,80,00,00,00,000	43,00,00,00,00,000	46,60,00,00,00,000
	China	12,00,00,00,00,000	13,20,00,00,00,000	14,50,00,00,00,000	16,40,00,00,00,000	19,30,00,00,00,000
	India	4,77,00,00,00,000	4,94,00,00,00,000	5,24,00,00,00,000	6,18,00,00,00,000	7,22,00,00,00,000
	South Korea	5,62,00,00,00,000	5,33,00,00,00,000	6,09,00,00,00,000	6,81,00,00,00,000	7,65,00,00,00,000
	Indonesia	1,65,00,00,00,000	1,60,00,00,00,000	1,96,00,00,00,000	2,35,00,00,00,000	2,57,00,00,00,000
	Thailand	1,23,00,00,00,000	1,16,00,00,00,000	1,27,00,00,00,000	1,43,00,00,00,000	1,61,00,00,00,000
	Hong Kong	1,72,00,00,00,000	1,69,00,00,00,000	1,66,00,00,00,000	1,61,00,00,00,000	1,69,00,00,00,000
	Malaysia	93,78,94,73,684	92,78,39,47,368	1,01,00,00,00,000	1,10,00,00,00,000	1,25,00,00,00,000
	Singapore	95,83,59,70,989	89,28,50,87,395	91,94,17,91,944	97,00,23,05,536	1,14,00,00,00,000
	Philippines	81,02,62,94,681	76,26,19,98,623	81,35,76,57,790	83,90,82,05,720	91,37,12,36,939
	Pakistan	73,95,23,74,970	72,30,97,38,921	72,30,68,20,396	83,24,48,01,093	97,97,77,66,198
	Kazakhstan	18,29,19,90,619	22,15,26,89,130	24,63,65,98,581	30,83,36,92,831	43,15,16,47,003
	Vietnam	33,64,00,85,728	35,29,13,49,277	37,94,79,04,054	42,71,70,72,778	49,42,41,07,710
	Bangladesh	47,12,49,25,462	46,98,78,42,847	47,57,11,30,071	51,91,36,61,485	56,56,07,44,012
	Sri Lanka	16,33,08,10,304	15,74,62,24,410	17,10,26,23,876	18,88,17,65,437	20,66,25,25,941
	Azerbaijan	5,27,26,17,196	5,70,76,18,247	6,23,60,24,951	7,27,57,66,111	8,68,05,11,918
	Uzbekistan	13,76,03,74,488	11,40,13,51,420	9,68,79,51,055	10,12,81,12,401	12,03,00,23,548
	Macao	6,10,17,94,939	6,51,42,71,488	7,00,80,26,415	7,92,63,73,572	10,25,83,24,479
	Turkmenistan	2,90,46,62,605	3,53,47,71,969	4,46,20,28,989	5,97,74,40,583	6,83,83,51,088
	Nepal	5,49,42,52,208	6,00,70,61,224	6,05,08,75,807	6,33,04,76,435	7,27,39,33,993
	Brunei	6,00,11,53,318	5,60,10,90,584	5,84,33,29,102	6,55,73,33,067	7,87,23,33,197
	Afghanistan	2,46,16,66,315	2,46,16,66,315	4,12,88,18,043	4,58,36,48,922	5,28,54,61,999

Region
☐ (All)
☐ Africa
☒ Asia
☐ Europe
☐ Middle East
☐ Oceania
☐ The Americas
SUM(GDP)
350M 16B

- Business Tax Rate:** Most countries in the middle East had low to moderate business tax rates, which were consistent throughout the years, except Yemen which had an unusually high business tax rate of 1.952% in the year 2005.

